Sham Satish Thakare

Computer Science Researcher · *Reliable and Adaptive Intelligent Systems*

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RESEARCH INTERESTS

Reliable and Adaptive Intelligent Systems: Uncertainty calibration in language-model post-training (RLVR credit assignment, reasoning complexity confounding), pre-hoc plasticity diagnostics for foundation model scheduling, multi-stage agent tool safety, failure-domain aware consensus quorum adaptation in C++20, and microservice causal observability.

EDUCATION

Anantrao Pawar College of Engineering and Research (APCOER), Pune, India **01/2020 – 06/2024**

Bachelor of Technology (B.Tech.) in Artificial Intelligence and Data Science | **Cumulative GPA: 8.70 / 10.00**

- **B.Tech. Research Thesis:** *Machine Learning Diagnostic Evaluation Protocols & Statistical Algorithms* (Advisers: Prof. Sagar Gawande & Prof. Sunil Thakare)
- **Mathematics & CS Coursework:** Linear Algebra, Multivariate Calculus, Probability, Machine Learning, Deep Learning, Reinforcement Learning, Distributed Systems, Algorithms.

RESEARCH EXPERIENCE

Independent Computer Science Researcher — *Reliable Adaptive Intelligent Systems* **06/2024 – Present**

Uncertainty-Weighted Credit Assignment in LM RLVR (Flagship Study) | 01/2026 – Present

- **PROBLEM:** Evaluated whether token predictive entropy validly measures reasoning uncertainty in GRPO post-training, or operates as a proxy for sequence length.
- **METHOD:** Formulated PyTorch diagnostic probes and a 5-way RL benchmark across GSM8K prompts (N=100) with Qwen2.5-0.5B-Instruct over N=3 seeds.
- **EVIDENCE:** Identified strong positive correlation between token entropy and completion length (r = 0.486, 95% CI [+0.318, +0.627]), collapsing to non-significance ($r_{partial}$ = -0.092, p = 0.365) after length controls; proved MC-dropout determinism (Var=0) on zero-dropout architectures; proved via negative controls zero observed online advantage (Cohen's d = 0.00, 80.00% Pass@1) over outcome-supervised GRPO.
- **CONTRIBUTION:** Released open-source PyTorch probing suite ear_grpo_reasoning and canonical research essay on RLVR diagnostic protocols.

CARLS: Foundation Model RL Plasticity Prediction | 02/2026 – Present

- **PROBLEM:** Addressed compute overhead in executing full policy-gradient RL loops to select optimal pre-trained LM checkpoints for post-training alignment.
- **METHOD & EVIDENCE:** Formulated gradient norm and feature drift probes across Pythia/Qwen series, predicting reward plasticity pre-hoc and saving 18.7% RL training FLOPs.
- **CONTRIBUTION:** Developed open-source RL plasticity benchmarking library adaptive-rl-forge.

QuorumShift: Failure-Domain Aware Consensus Quorum Adaptation | 06/2024 – 01/2026

- **PROBLEM:** Eliminated tail latency spikes and availability degradation in Raft consensus clusters under asymmetric network partitioning.
- **METHOD & EVIDENCE:** Designed C++20 dynamic quorum adaptation adjusting voting thresholds via real-time latency profiles, reducing 99th percentile write latency.
- **CONTRIBUTION:** Built high-performance C++20 open-source library quorumshift.

Undergraduate Researcher — APCOER Pune | *Advisers: Prof. Sagar Gawande & Prof. Sunil Thakare* **02/2023 – 05/2024**

- Formulated machine learning diagnostic evaluation frameworks and statistical algorithms for high-dimensional data evaluation.

SELECTED PUBLICATIONS & MANUSCRIPTS

- **[Flagship Manuscript] Thakare, S. S.** "When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in LLMs." *Working Paper / Research Note*, 2026. [PDF Paper] [Code] [Research Essay]
  Exposes length confounding (r = 0.486, N=100) in token entropy, proves MC-dropout determinism on zero-dropout LLMs, and proves via negative controls zero online advantage (d = 0.00) over outcome-supervised GRPO.
- **[RL Plasticity] Thakare, S. S.** "Predicting Reinforcement-Learning Plasticity of Intermediate Language-Model Checkpoints." *Working Paper*, 2026. [Code]
  Formulates representation geometry probes across Pythia/Qwen series, predicting reward plasticity pre-hoc and saving 18.7% FLOPs.
- **[Distributed Consensus] Thakare, S. S.** "AdaptiveReplica: Dynamic Quorum Adaptation in Distributed Consensus." *Working Paper*, 2026. [Code]
  C++20 dynamic quorum adaptation engine for Raft consensus clusters eliminating tail latency under asymmetric network partitioning.
- **[Error Recovery] Thakare, S. S.** "recovery_eval: State-Matched Evaluation of Recovery in LM Reasoning." *Working Paper*, 2026. [PDF Paper] [Code]

• **[Agent Safety] Thakare, S. S.** "MediRush-SafeAgent: Multi-Stage Interception Architecture for Autonomous Agent Safety." *Research Project*, 2026. [Code]

• **[Cloud Systems] Thakare, S. S.** "TraceMind: Graph-Constrained Causal Reasoning for Microservice Root-Cause Localization." *Working Paper*, 2026. [Code]

• **[Hardware Enclaves] Thakare, S. S.** "EnclaveShield: Zero-Knowledge Memory Attestation for Hardware Enclaves." *Working Paper*, 2026. [Code]

PROFESSIONAL EXPERIENCE

Capital One | *Pre-Doctoral Researcher* (Research Mentor: Yuri Shelke, Distinguished Engineer) **06/2024 – Present**

• Researched distributed consensus protocols and C++20 dynamic quorum adaptation for Raft clusters under asymmetric network partitioning.

• Formulated moving-window latency profiling algorithms and multithreaded execution telemetry, reducing p99 write latency by 41.3%.

TEACHING EXPERIENCE

APCOER Pune (Dept. of AI & DS), Pune, India | *Teaching Assistant* **06/2023 – 05/2024**

• Provided instructional support and doubt resolution for *Data Structures*, *Algorithms*, and *Foundations of Machine Learning*.

• Conducted lab sessions, graded programming assignments, and developed automated grading scripts in Python and C++.

HONORS & AWARDS

• **Gold Medal in Mathematics** — Highest Academic Standing in Mathematics & Quantitative Engineering Curricula, APCOER Pune (SPPU), 2024.

• **APCOER Academic Excellence Distinction** — Department of AI & Data Science (Cumulative GPA: 8.70 / 10.00; Final Year SGPA: 9.53 / 10.00), 2024.

• **Gold Medal, Athletics (Sprint / Distance Running)** — Inter-College Athletic Meet, 2023.

LEADERSHIP & COMMUNITY SERVICE

• **President, Social Volunteer Club** — APCOER Pune (2023–2024). Elected student head of university community outreach. Directed district-wide blood donation drives, educational mentoring for underprivileged youth, campus sustainability campaigns, and community welfare programs across Pune.

OPEN SOURCE RESEARCH SOFTWARE

• **ear_grpo_reasoning**: PyTorch probing suite for token predictive entropy & GRPO advantage scaling.

• **adaptive-rl-forge**: Diagnostic probe library for foundation model checkpoint reward plasticity.

• **quorumshift**: High-performance C++20 dynamic quorum adaptation for Raft consensus clusters.

• **medirush-safeagent**: Multi-stage runtime policy interception architecture for agent tool safety.

• **recovery_eval**: State-contrast continuation evaluation framework for neural reasoning error recovery.

TECHNICAL SKILLS

• **Languages & Frameworks:** Python, C++ (C++20), PyTorch, Hugging Face, SQL, Java, JavaScript/TypeScript, POSIX.

• **Systems & Methods:** C++20 Multithreading, Raft Consensus, OpenTelemetry, Docker, Linux, PyTest, Statistical Controls, Multi-Seed Probing.